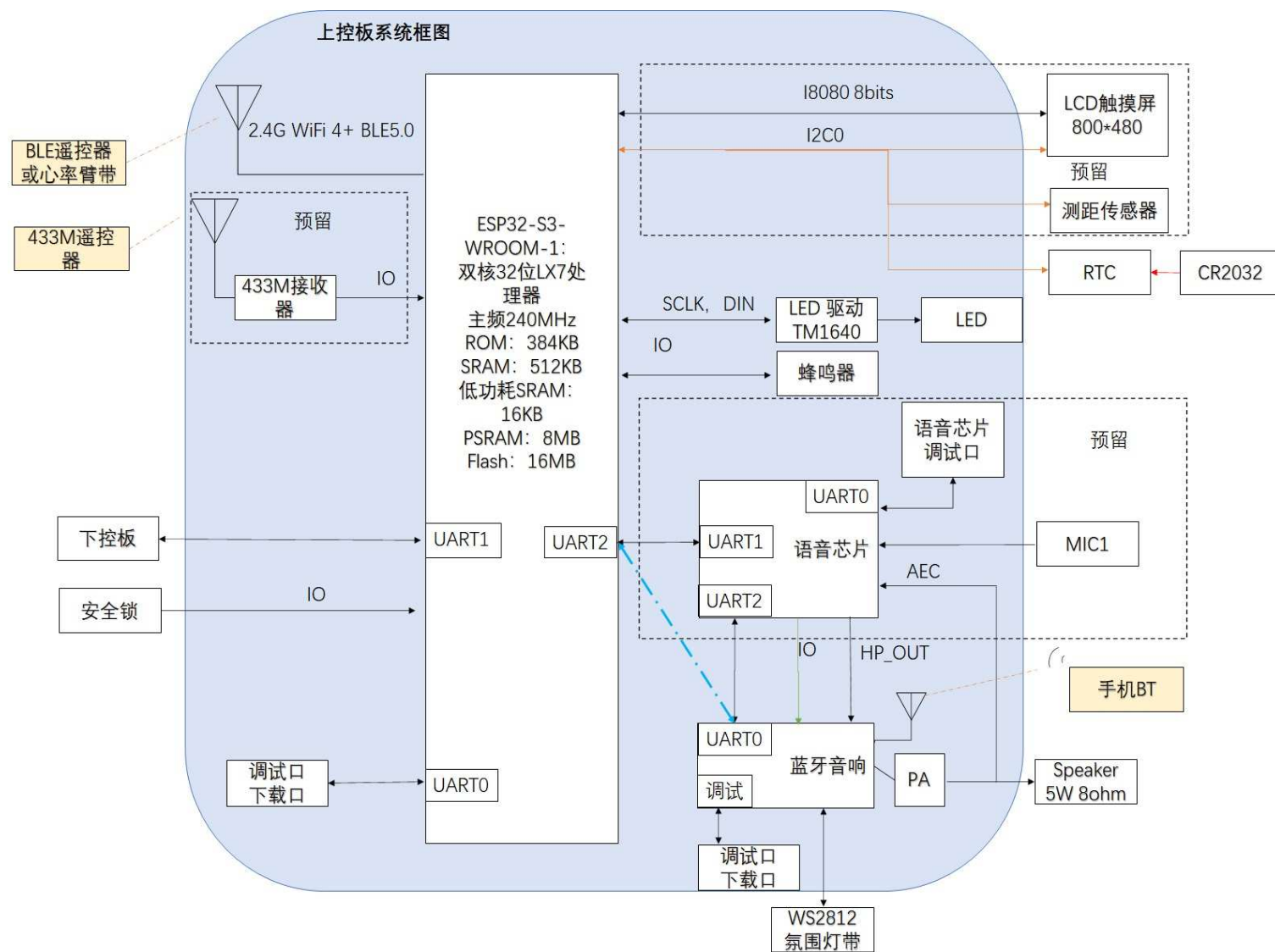


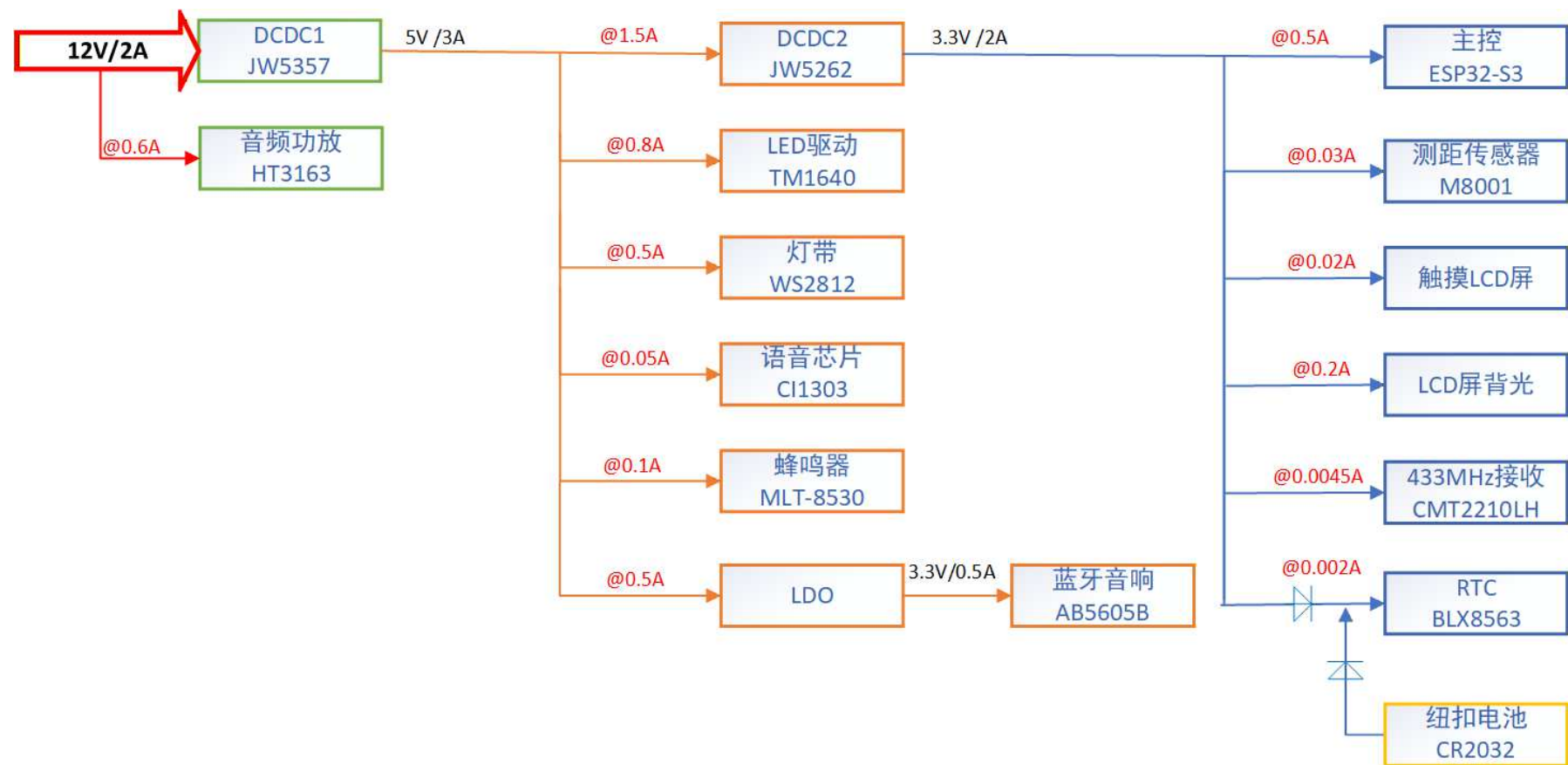
BLOCK



V0.1

Zhejiang Mairuike Technology Co., Ltd.			
Design Name			
T28_Upper Control Board			
Size	Page Name	Rev	
A3	BLOCK	<V01>	
Date	Friday, June 12, 2026	Sheet	1 of 10

POWER TREE



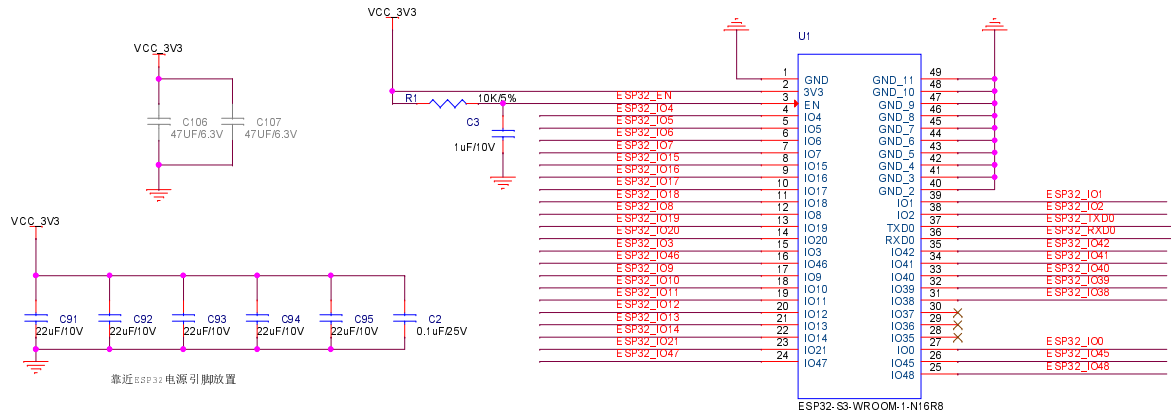
V0.1

 Zhejiang Mairuike Technology Co., Ltd.			
Design Name	T28_Upper Control Board		
Size A3	Page Name	POWER TREE	Rev <V01>
Date	Friday, June 12, 2026	Sheet 2 of	10

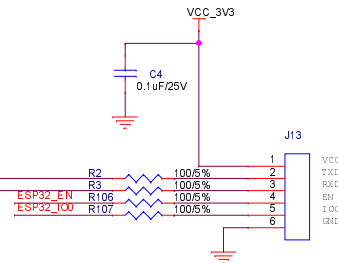
分页标识

LCD_CS_n	<<	ESP32_IO48	TP19
LCD_RS	<<	ESP32_IO45	TP20
LCD_WR_n	<<	ESP32_IO0	TP21
LCD_RD_n	<<	ESP32_IO38	TP22
LCD_RST_n	<<	ESP32_IO39	TP23
LCD_D0	<<	ESP32_IO10	TP24
LCD_D1	<<	ESP32_IO9	TP25
LCD_D2	<<	ESP32_IO11	TP26
LCD_D3	<<	ESP32_IO12	TP27
LCD_D4	<<	ESP32_IO13	TP28
LCD_D5	<<	ESP32_IO14	TP29
LCD_D6	<<	ESP32_IO21	TP30
LCD_D7	<<	ESP32_IO47	TP31
M8001_INT	<<	M8001_INT	TP32
TP_INT	<<	TP_INT	TP33
M8001_RST	<<	M8001_RST	TP34
TP_RST	<<	TP_RST	TP35
WS2812_DIN	<<	WS2812_DIN	TP36
PA_CTRL	<<	ESP32_IO6	TP37
I2C_SDA	<<	ESP32_IO6	TP38
I2C_CLK	<<	ESP32_IO7	TP39
ESP_TX1	<<	ESP32_IO17	TP40
ESP_RX1	<<	ESP32_IO18	TP41
TM1640_LED_DIN	<<	TM1640_LED_DIN	TP42
ESP_TX2	<<	ESP32_IO19	TP43
ESP_RX2	<<	ESP32_IO20	TP44
TM1640_LED_SCK	<<	TM1640_LED_SCK	TP45
433M_DIN	<<	ESP32_IO42	TP46
SAFETY_IN	<<	ESP32_IO41	TP47
BL_PWM	<<	ESP32_IO40	TP48
PWR_KEY	<<	ESP32_IO1	TP49
BZ_OUT	<<	ESP32_IO2	TP50

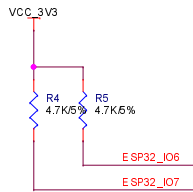
ESP32-S3



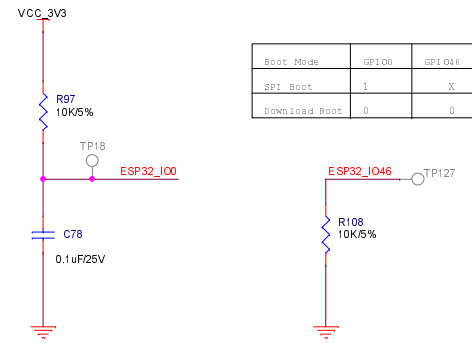
烧录调试口



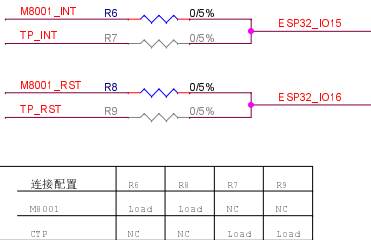
I2C 上拉电阻



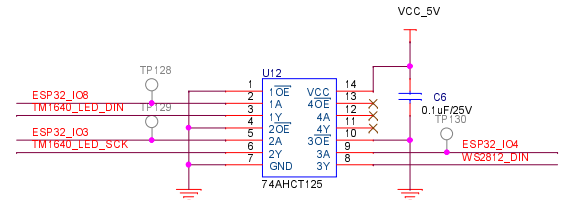
BOOT



引脚复用

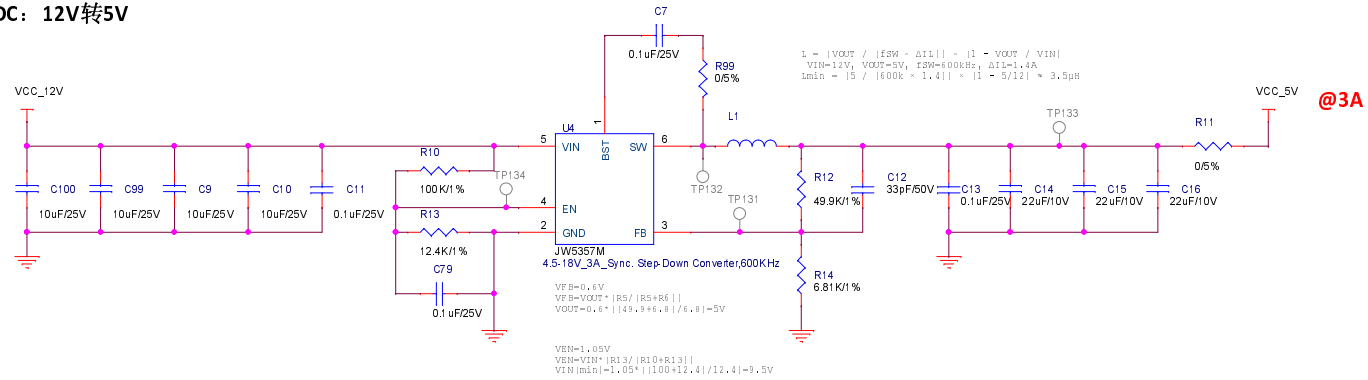


电平转换

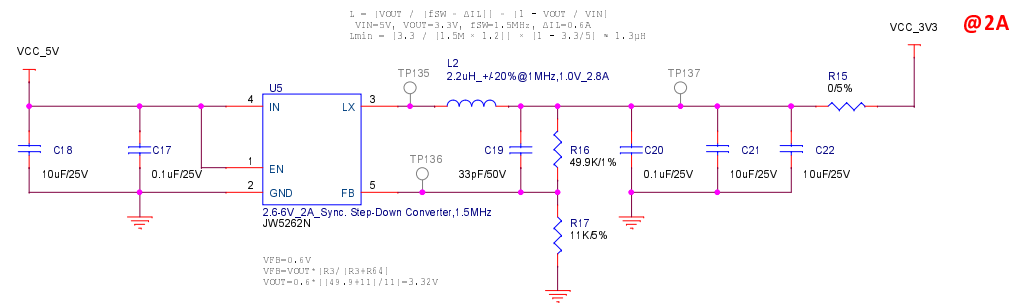


分页标识

DCDC: 12V转5V



DCDC: 5V转3V3



V0.1

Zhejiang Mairuike Technology Co., Ltd.			
Design Name			
T28_Upper Control Board			
Size	Page Name	Rev	
A3	POWER	<V01>	
Date	Friday, June 12, 2026	Sheet	4 of 10

分页标识



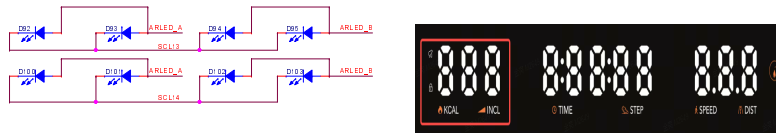
WiFi

静音

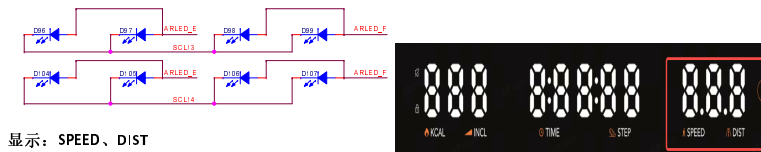
BT

童锁

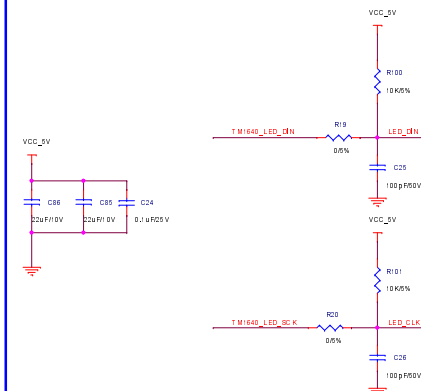
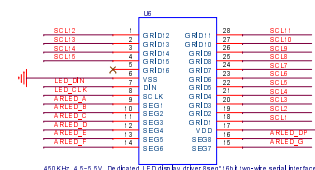
显示: KCAL、INCL



显示: SPEED、DIST



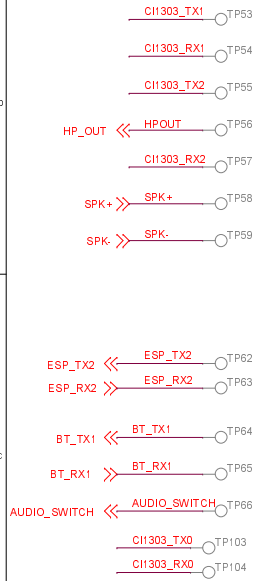
LED 断码驱动



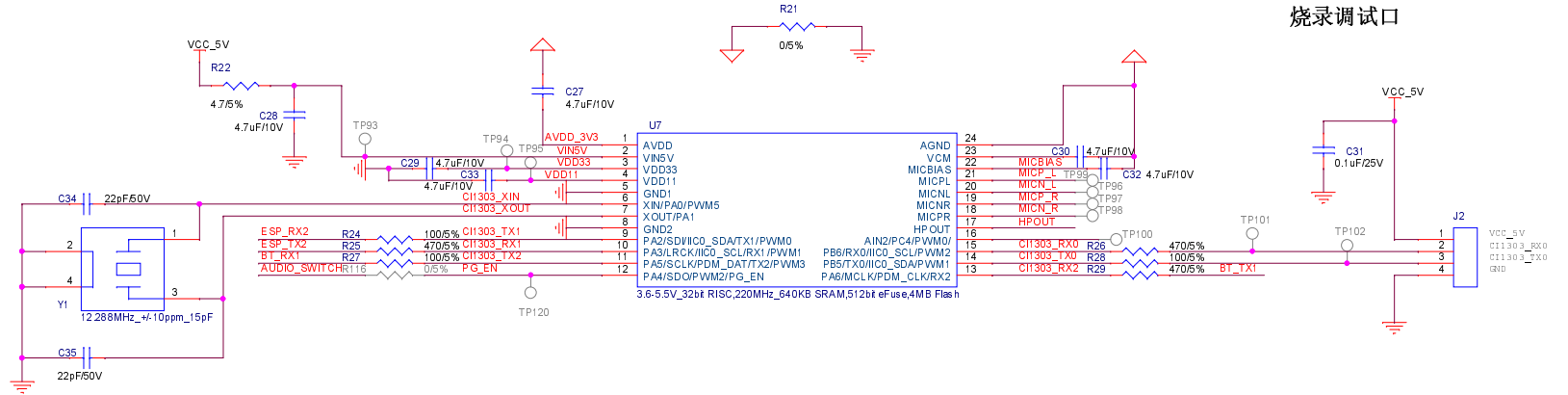
显示: TIME、STEP



分页标识

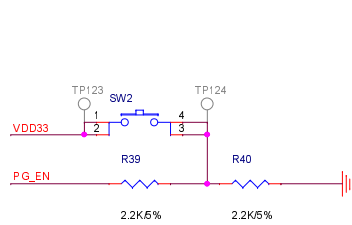


CI1303



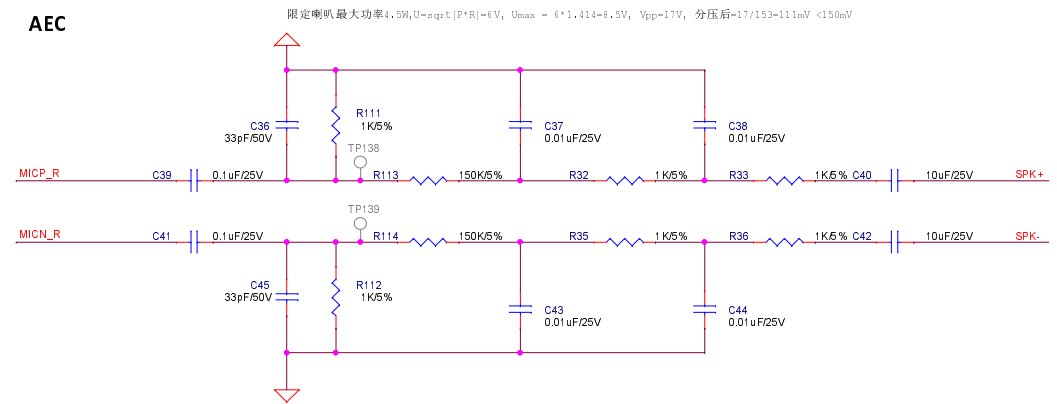
烧录调试口

BOOT按键

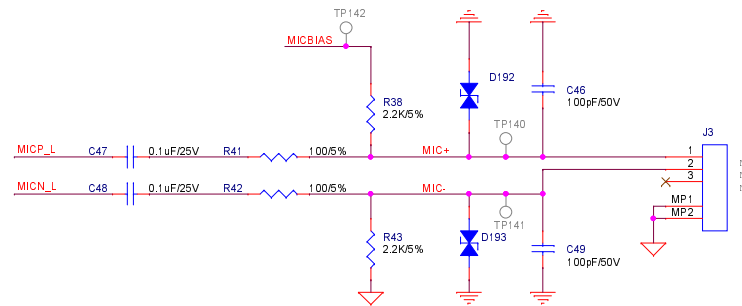


Boot Mode	P_G_EN
升级模式	1
普通模式	0

AEC



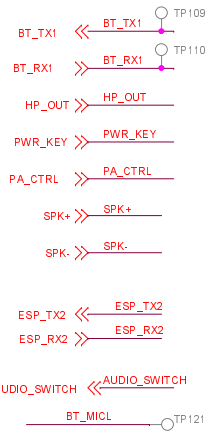
麦克风



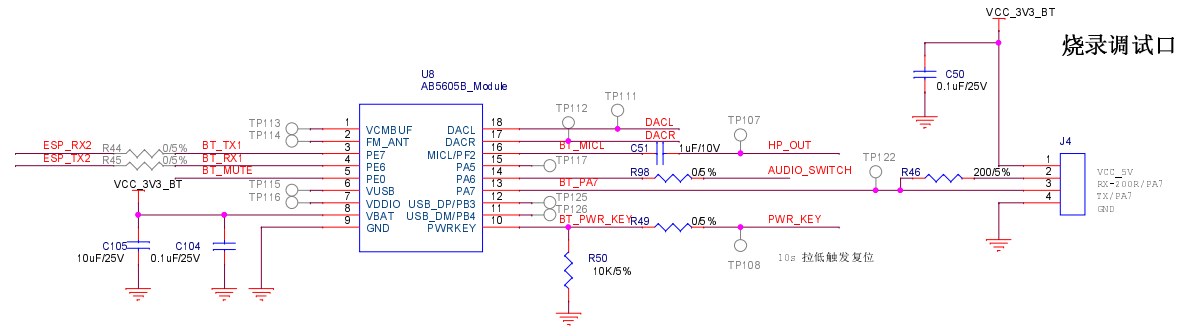
V0.1

Zhejiang Mairuike Technology Co., Ltd.			
Design Name	T28_Upper Control Board		
Size A3	Page Name	AUDIO	
Date	Friday, June 12, 2026	Sheet	6 of 10

分页标识

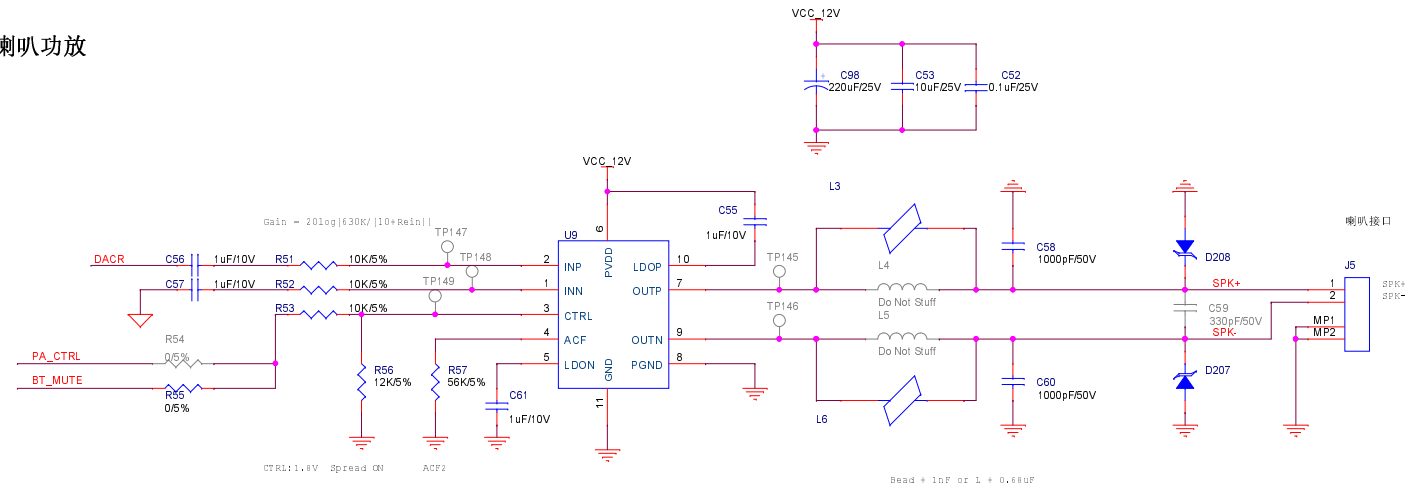


AB5605

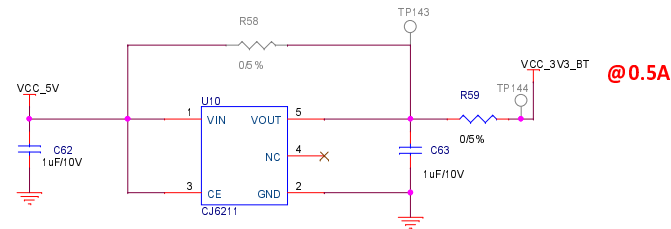


烧录调试口

喇叭功放



LDO: 5V转3.3V



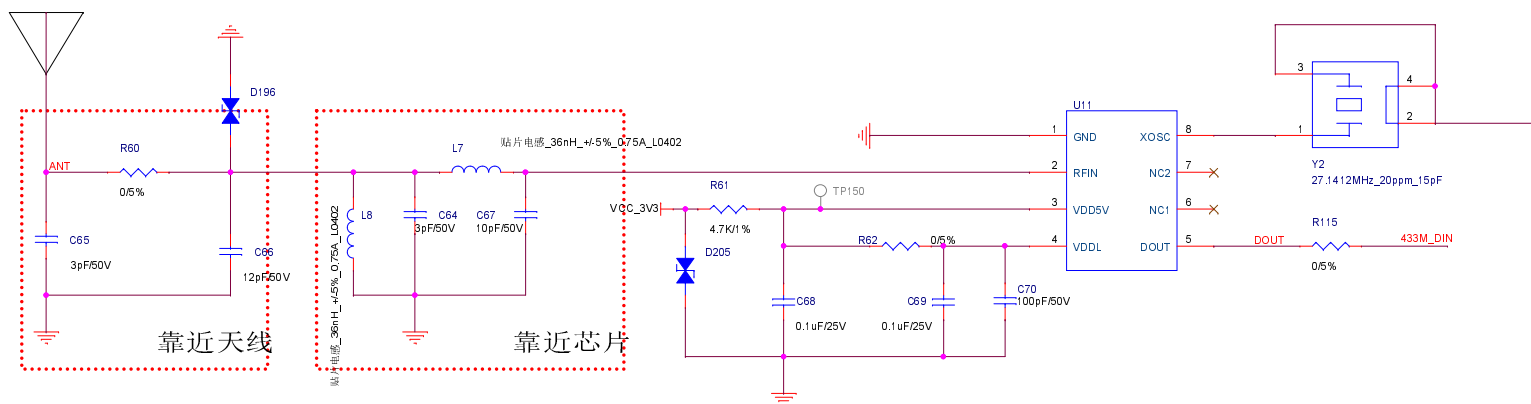
V0.1

			
Zhejiang Mairuike Technology Co., Ltd.			
Design Name T28_Upper Control Board			
Size A3	Page Name	BT MODULE	
Date	Friday, June 12, 2026	Sheet 7	of 10

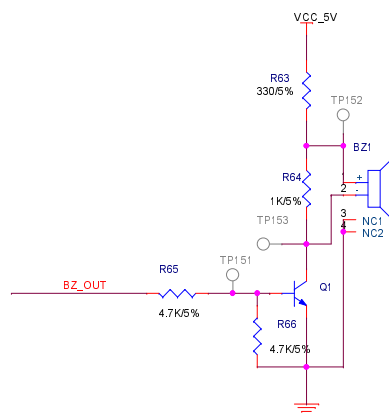
分页标识

BZ_OUT >> BZ_OUT
433M_DIN << 433M_DIN

433M接收



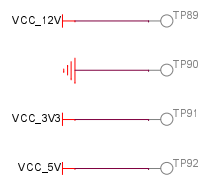
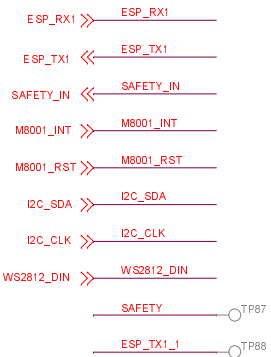
蜂鸣器



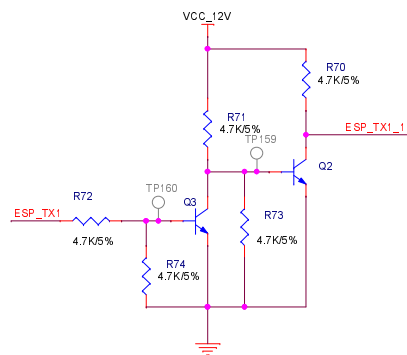
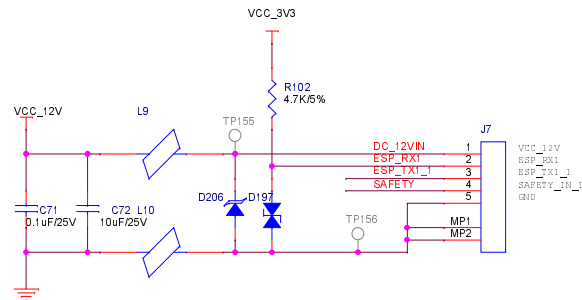
V0.1

		Zhejiang Mairuike Technology Co., Ltd.	
Design Name		T28_Upper Control Board	
Size A3	Page Name RF	Rev <V01>	
Date	Friday, June 12, 2026	Sheet	8 of 10

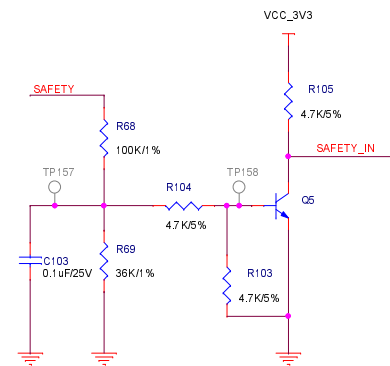
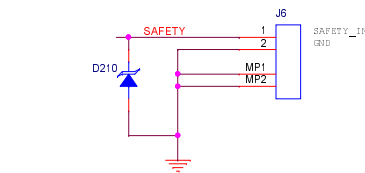
分页标识



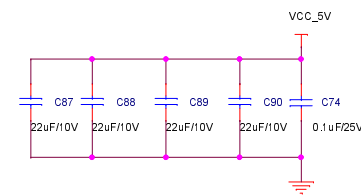
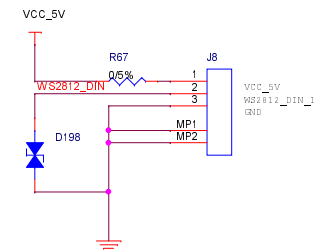
下控板接口



安全锁接口

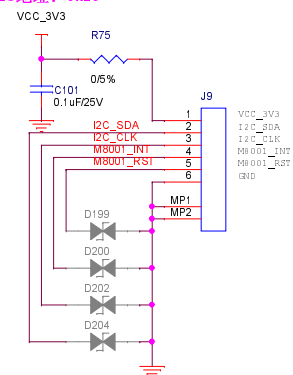


灯带接口



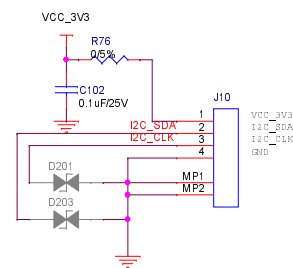
测距传感器接口

I2C地址: 0x10



RTC小板接口

I2C地址: 0x51



V0.1

			
Zhejiang Mairuike Technology Co., Ltd.			
Design Name		T28_Upper Control Board	
Size	Page Name	Rev	
A3	INTERFACE	<V01>	
Date:	Friday, June 12, 2026	Sheet	9 of 10

D	LCD_CS_n	>>>	LCD_CS_n
	LCD_RS	>>>	LCD_RS
	LCD_WR_n	>>>	LCD_WR_n
	LCD_RD_n	>>>	LCD_RD_n
	LCD_RST_n	>>>	LCD_RST_n
	LCD_D0	>>>	LCD_D0
	LCD_D1	>>>	LCD_D1
	LCD_D2	>>>	LCD_D2
	LCD_D3	>>>	LCD_D3
	LCD_D4	>>>	LCD_D4
C	LCD_D5	>>>	LCD_D5
	LCD_D6	>>>	LCD_D6
	LCD_D7	>>>	LCD_D7
	BL_PWM	>>>	BL_PWM
	TP_INT	>>>	TP_INT
	TP_RST	>>>	TP_RST
	I2C_SDA	>>>	I2C_SDA
	I2C_CLK	>>>	I2C_CLK

I2C地址: 0x5D or 0x14

